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A Hierarchical Mixture-of-Experts Framework (HiMoE-F) for Real-Time Packaging Label Defect Detection

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Abstract Product packaging plays an important role as a protection, communication, and marketing tool across industrial sectors. Packaging defects can affect product quality, consumer trust, and brand credibility. Industrial packaging lines require fast and reliable inspection, but medical packaging label inspection remains challenging because of class-wise variability and defect-scale variation. This paper proposes a Hierarchical Mixture-of-Experts Framework (HiMoE-F) for real-time packaging label defect detection. Unlike end-to-end trainable Mixture-of-Experts architectures, HiMoE-F uses independently trained models with rule-based hierarchical routing and fusion to balance recall and latency for industrial deployment. We use a high-resolution dataset with 2,250 images and ten classes (nine defect types and one Good Sample). First, multiple real-time models are trained and tuned using Optuna. Next, a base expert de-

tector and class-specialized expert detector are selected using strategy that ranks candidate models for each class based a weighted sum model that combines recall and inference latency. In the final deployment pipeline, YOLOv8m runs on every frame as the base expert detector, while YOLOv10m is executed conditionally as a specialist for Barcode Defect. Detections are merged using class-wise non-maximum suppression, followed by a PASS / REJECT / UNKNOWN decision layer. On the test set, HiMoE-F achieves 0.973 recall, 0.985 precision, 0.962 mAP@0.5, and 0.84 mAP@0.5:0.95, while maintaining 43.34 ms average inference latency per image. The results show that HiMoE-F improves recall-oriented deployment reliability while preserving real-time feasibility.

Keywords Industrial Inspection · Packaging Label · Object Detection · Hierarchical Gating · Class-Specialized Experts · Recall-Oriented Optimization

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1 Introduction

Product packaging serves as an essential tool for protection and communication, particularly in the food and medical sectors where packaging errors can directly affect safety, regulatory compliance, and user trust [21]. To ensure efficient quality control (QC) during production, the industry has transitioned from manual inspection which is limited by worker fatigue and inconsistency to automated, data driven systems. Driven by the growth of Industry 4.0, computer vision and deep learning have become increasingly critical for these automated quality inspections [5], providing necessary reliability and efficiency without the limitations of human visual processes.

Table 1: Correspondence between HiMoE-F components and specific challenges in packaging-label defect detection.

Modules	Scale diversity	Class-wise variability
Base Expert Detector (generalist)	✓	✗
Class-Specialized Expert Detectors (specialists)	✓	✓
Base expert detector → class-specialized expert detector gating	✗	✓

Deep learning has substantially improved industrial inspection because it learns features directly from images and can handle complex textures [14]. Real-time object detection has advanced rapidly with modern model families such as YOLO variants and transformer-based models [4]. However, as industrial automation continues to advance, the demand for packaging defect detection with both high accuracy and low inference latency is increasing. According to the industrial partner’s deployment requirements, the target is 99% recall together with end-to-end processing latency below 100 ms per image. Improving both detection accuracy and latency is therefore critical. Unlike general object detection tasks, medical packaging label inspection faces several specific challenges such as substantial variation in defect size, shape, appearance and severe class-wise variability with limited samples for rare sample images.

Many studies focus on a single packaging component, such as only a barcode, only a blister pocket, or only printing defects, rather than a unified multi-class label inspection setting. Some works report strong accuracy, but provide limited analysis of latency stability, which is essential for production deployment. To address these limitations, we propose HiMoE-F, a deployment-oriented hierarchical expert framework that combines recall-oriented tuning, class-specialized expert selection, trigger-based specialist execution, class-wise fusion with non-maximum suppression (NMS), and a PASS / REJECT / UNKNOWN decision layer. The relationship between the main components and the targeted deployment challenges is summarized in Table 1.

To validate the effectiveness of the proposed method, we conducted experiments on the Medical Packaging Dataset (MPD), which reflects key challenges such as defect-pattern variation, class-wise variability, limited benchmarks, and real-time constraints. The results show

that HiMoE-F improves recall compared with all evaluated single-model baselines while maintaining practical deployment inference latency.

The main contributions of this study are fourfold. First, we propose HiMoE-F, a hierarchical expert framework for real-time defect detection in medical packaging labels. Second, we introduce a class-specialized expert selection strategy that ranks candidate models for each class using a weighted sum model that combines recall and inference latency [19]. Third, we design a practical deployment decision layer that assigns each sample to PASS, REJECT, or UNKNOWN based on the final detection results. Finally, we evaluate the proposed framework on the Medical Packaging Dataset (MPD) and compare its performance with several baseline object detection models in terms of precision, recall, mAP, and inference latency.

2 Related Work

2.1 Deep Learning for Packaging Label Inspection

Deep learning architectures have been increasingly applied to diverse industrial inspection tasks, ranging from identifying general label and printing defects to addressing barcode and QR code issues and pharmaceutical packaging quality control [13]. Research has successfully deployed various models, such as the YOLO series and EfficientDet, for detecting damaged logistics boxes [24], imperfections in thermal-transfer printed labels [13], and the integrity of safety closures in food packaging [8]. In pharmaceutical contexts, specialized frameworks such as PharmaNet Deep and CBS-YOLOv8 are used to inspect blister packs for tablet-level defects such as cracks, color mismatches, and foreign particles after packaging [22].

The packaging domain presents significant technical challenges for visual inspection systems. Defects are often subtle or very small, such as micro-cracks or tiny black spots, and can easily be confused with the background texture of the packaging material [12]. Smear-like ambiguities and low contrast between the defect and the intricate design of printed materials further complicate detection [16]. In addition, industrial environments face substantial domain shift across different packaging designs, which frequently change, and traditional machine-vision systems lack the flexibility to adapt without extensive redesign of hand-crafted feature extractors [10]. This challenge is compounded by class-wise variability with limited samples for rare defect classes, leading to a long-tailed data distribution that can cause models to overlook rare defect modes [12].

Despite these advances, the existing literature often focuses on narrow subtasks. Many studies address only

a single component of the problem, such as barcode inspection, blister-pocket inspection, or specific printing artifacts. As a result, there is limited work on a unified framework capable of handling the full diversity of label defects encountered in real production environments under the same real-time constraints.

2.2 Real-time object detection for industrial inspection

One-stage models, especially the YOLO (You Only Look Once) series and SSD, are highly popular in industrial contexts because they achieve the inference speeds required for real-time monitoring [20]. Unlike two-stage models, these models perform object localization and classification in a single forward pass, which significantly reduces computational overhead and allows them to be integrated directly into fast-paced production lines [17].

To enhance detection capability, the DETR family and Vision Transformers (ViTs) have been introduced to bring global reasoning to inspection tasks [15]. These architectures are designed to capture fine-grained spatial details and learn long-range dependencies between image patches, regardless of spatial distance. Although transformers were traditionally considered more computationally intensive, real-time variants are increasingly being developed to take advantage of their contextual modeling ability.

However, even strong models frequently show uneven per class behavior when applied to industrial data [23]. A model’s accuracy often fluctuates because of class-wise variability with limited samples for rare defect classes, where it may become proficient at identifying common defects but struggle with rarer ones. Performance is also often inconsistent across different defect types; while structured elements are easier to detect, subtle defects such as faint misprints and small contaminations often lack distinctive visual evidence and therefore lead to missed detections [15].

Existing real-time models are typically deployed as a single universal model, which can perform inconsistently across defect types that range from structured patterns such as text or QR elements to visually confusing patterns such as smears.

2.3 Mixture-of-Experts and class specialization for detection

The concept behind HiMoE-F is related to the broader idea of activating specialists conditionally based on the input data [6]. Hierarchical expert systems can establish relationships between coarse and fine categories so that specialized branches focus on more difficult cases

while still benefiting from broader contextual information. In modern vision tasks, MoE-style routing and dedicated specialist branches can improve performance by allocating model capacity to hard cases [18]. These architectures often incorporate defect-guided attention mechanisms or specialized feature reconstructors to focus computational resources on informative image regions, particularly when dealing with subtle or partially occluded defects.

Despite these advances, many MoE and specialized detection studies prioritize generic benchmark accuracy such as mean Average Precision (mAP) [1]. In contrast, industrial quality control requires a recall-dominant approach, where the primary objective is to identify as many true defects as possible to prevent faulty products from reaching downstream processes [3]. In addition, many end-to-end MoE architectures rely on learned routing and more complex training procedures, which may complicate deployment in industrial environments.

In this work, HiMoE-F is positioned as a deployment-oriented hierarchical expert framework rather than an end-to-end trainable MoE detector. It uses independently trained models together with rule-based expert selection, trigger-based execution, and lightweight fusion to improve recall while maintaining practical deployment control.

2.4 Conditional computation, fusion, and deployment logic

Conditional execution is frequently used in industrial inspection to improve efficiency by running additional computation only when specific criteria are met. For instance, some frameworks use a twice-matching strategy in which intensive feature analysis is performed only on latent defect candidates identified in a preliminary pass, thereby reducing the overall computational burden [11]. Similarly, modular architectures can conditionally activate specialized branches to refine detections for hard-to-classify defect types [18].

Multi-model fusion techniques such as NMS are also important for merging outputs from different detection heads or models. In industrial settings, these fusion strategies must remain computationally light to preserve real-time performance. Advanced variants, such as the multi-task NMS used in DML Mask R-CNN, allow systems to handle dependent multi-label outputs by considering combinations of confidence and severity scores during filtering [11, 18].

Open-set detection and uncertainty estimation are related to the broader need for conservative handling of visually confusing deployment cases. Modern mod-

els such as PharmaNet Deep incorporate uncertainty-aware detection heads that model uncertainty alongside class probabilities [22]. However, many studies treat uncertainty as a separate academic metric rather than integrating it into a direct deployment decision. In production environments, the output must ultimately be translated into actionable results such as pass or reject commands, often for interaction with a Programmable Logic Controller (PLC) [24, 22]. Some systems therefore propose hybrid inspection pipelines in which deep learning handles standard cases while uncertain items are routed for manual review [20].

Prior works on conditional inference and detector fusion rarely combine class-specialized routing, real-time latency reporting, and a practical PASS / REJECT / UNKNOWN decision layer for deployment. To address this gap, we propose a base-expert-detector and class-specialized-expert-detector HiMoE-F framework that combines recall-oriented expert selection, trigger-based specialist execution, efficient fusion, and deployment decision logic.

3 Methodology

This work targets real-time defect detection for medical packaging labels. Packaging defects are very diverse in shape and size. Some defects are structured and easy to detect, but others are weak, visually confusing, and difficult to localize. Because of this, a single model often performs not consistently across defect types. To address this limitation, we propose HiMoE-F, a system that uses one general model as a base expert detector and activates class-specialized expert detector only when needed. This design follows the general idea of hierarchical expert systems, in which different models focus on different parts of the task [9].

The overall system consists of three stages Optuna-based hyperparameter tuning for each candidate model, where each trial trains and validates a model and returns validation recall to the study, class-specialized expert selection using a weighted score based on normalized recall and normalized latency on the held-out expert selection split to assign the best model to each class, and a deployment HiMoE-F, in which a base expert detector runs on every frame and class-specialized expert detector are activated using trigger policies, followed by fusion with class-wise NMS and a decision layer.

3.1 Stage 1 Optuna model tuning

We first train a set of candidate real-time object models. Each model is tuned using automated hyperparam-

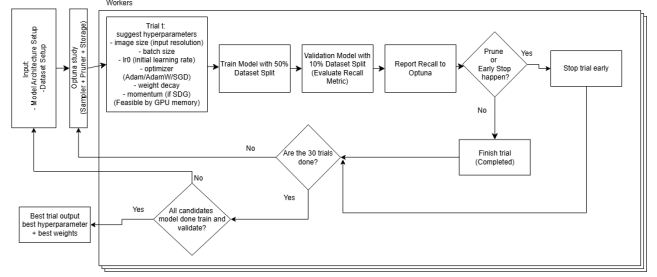


Fig. 1: Optuna-based recall-oriented tuning workflow (Stage 1)

ter optimization. We use Optuna to search for model-specific training configurations because manual tuning is slow and can lead to unfair comparisons across architectures [2]. The tuning is performed independently for each candidate model so that each model is optimized under the same tuning strategy before evaluation and selection.

Figure 1 illustrates the Optuna tuning workflow used in this study. For each candidate model, we define a separate Optuna study and run 30 trials. In each trial, Optuna proposes a hyperparameter configuration and launches one complete training run using the Ultralytics training pipeline. The tuned hyperparameters are the same across models included input image size, batch size, initial learning rate, optimizer type, weight decay, and momentum when SGD is used. Batch size and input resolution are constrained by GPU memory, so the tuning process considers only possible combinations. This keeps the search practical for high-resolution industrial images and avoids out-of-memory failures.

We set recall as the primary tuning objective because missed defects are more costly than false alarms in industrial inspection. In each trial, after training, the model is evaluated on the validation split 10% and the resulting recall is returned to Optuna as the trial score. Optuna uses this score to guide subsequent sampling toward better configurations. To reduce unnecessary computation, training is limited to 100 epochs with early stopping patience of 10 epochs, and Optuna pruning can stop low-performing trials before completing the full training budget. This follows common practice in efficient hyperparameter optimization for deep learning, where pruning and early stopping improve resource efficiency [7].

At the end of stage 1, the best-performing trial is selected for each model architecture. We retain two outputs per model including the best hyperparameter configuration and the trained weights from the best trial. These tuned models are then used as fixed candidates in stage 2, where we perform per class evaluation and

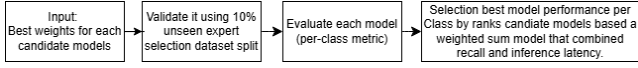


Fig. 2: Class-specialized expert selection workflow (Stage 2)

assign class-specialized expert detector. This separation ensures that the expert selection stage compares models that have already been tuned in a consistent, recall-oriented manner.

3.2 Stage 2 Class-specialized expert selection

Stage 2 converts those tuned models into a class-specialized expert set. The key idea is that packaging-label defects are diverse. Some defect types have clear structure, while others have weak boundaries, small regions, or high similarity to background patterns. As a result, a model that performs well overall may still be weak for a specific class. Stage 2 therefore performs per class evaluation across all tuned models and assigns the most suitable model to each class.

Figure 2 summarizes the class-specialized expert selection process. First, all tuned models are evaluated using the dedicated expert selection split, which is separate from both the validation split used for hyperparameter tuning and the final test split used for reporting locked pipeline performance. From each model, we extract the relevant metrics, then ranks the candidate models based a weighted sum model that combined recall and inference latency.

The output of stage 2 is an expert mapping that defines included a base expert detector that provides strong general performance across most classes, and one or more class-specialized expert detector assigned to difficult classes.

3.3 Stage 3 HiMoE-F inference pipeline

This stage is the deployment core of the proposed framework. It converts the stage 2 expert mapping into a real-time inspection pipeline that runs on every input frame. The pipeline is hierarchical and sequential, meaning that the base expert detector runs first and class-specialized expert detector are executed only when triggered by the base expert detector. This avoids running all models in parallel and helps keep latency stable for production use. Figure 3 illustrates the full inference flow implemented in our deployment code.

The framework begins with the model expert setup, in which the base expert detector and the class-specialized expert detector are configured before inference starts.

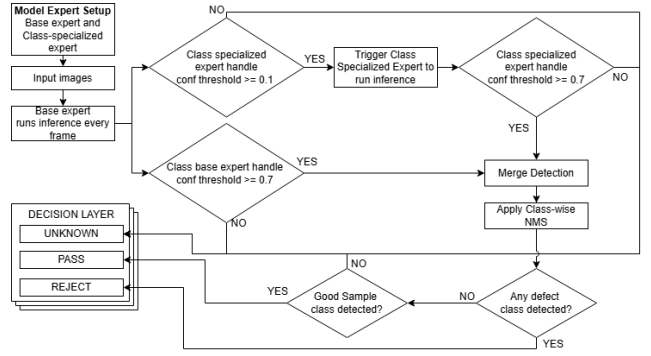


Fig. 3: HiMoE-F pipeline (Stage 3)

This setup defines the class ID and class-name mapping, the base expert detector parameters, and the class-specialized expert detector parameters, including weights, inference image size, confidence threshold, trigger threshold, and run policy. This design makes the framework reproducible and easy to extend, since additional experts can be added by updating the setup file without changing the overall pipeline logic.

For each input image, the base expert detector runs first and performs inference on every frame. In the final configuration of this study, the base expert detector is YOLOv8m. The base expert output is then separated into two paths according to the handling strategy of each class. For class-specialized-expert-handled classes, the base expert detections are checked against a trigger confidence threshold of 0.1. If the confidence score satisfies this threshold, the corresponding class-specialized expert is activated and runs inference on the same image. In this study, Barcode Defect is handled in this way, so a barcode detection from the base expert is used only to trigger the barcode specialist rather than being accepted directly as the final output.

After the class-specialized expert runs, its detections are evaluated using a stricter acceptance confidence threshold of 0.7. Only detections that satisfy this threshold are forwarded to the Merge Detection stage. In parallel, detections belonging to base-expert-handled classes are checked using the base expert acceptance threshold of 0.7. If the confidence score satisfies this threshold, the detection is also forwarded directly to the Merge Detection stage. Detections that do not satisfy the required threshold are discarded.

The accepted detections from both the base expert and the class-specialized expert are then combined in the Merge Detection module. Because multiple detections may overlap spatially, the merged outputs are refined using class-wise non-maximum suppression (NMS). This post-processing step removes redundant boxes within the same class while preserving detections from different

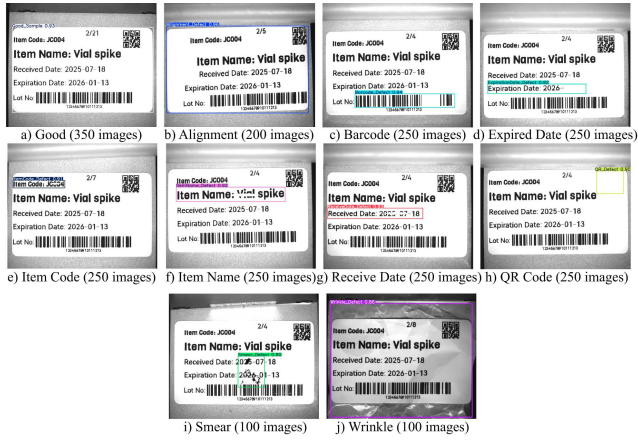


Fig. 4: Examples of packaging label defect classes in the MPD dataset

classes, thereby producing the final detection set for the input image.

Finally, the framework applies the decision layer to convert the final detections into a deployment-oriented output. First, it checks whether any defect class is detected. If at least one defect is present, the sample is assigned as REJECT. If no defect class is detected, the framework then checks whether the Good Sample class is present. If the Good Sample class is detected, the sample is assigned as PASS. Otherwise, if neither a defect class nor the Good Sample class is detected, the sample is assigned as UNKNOWN.

4 Experiments and Results

4.1 Experiment Dataset

In this study, to evaluate the proposed HiMoE-F framework for medical packaging label inspection, we used one dataset, namely the Medical Packaging Dataset (MPD), provided by Greatch Integration (M) Sdn. Bhd. as an industry partner for this project. The dataset consists of 2,250 high-resolution images, annotated with a total of 2,277 object instances. All images are stored in bitmap format with a fixed resolution of 2448×2048 pixels, ensuring fine-grained visual detail suitable for industrial inspection tasks. To provide an intuitive understanding of the dataset, Figure 4 shows example images for each class, while Figure 5 presents the class distribution across defect categories, since class-wise variability strongly influences recall for rare and difficult defects.

The MPD dataset covers ten classes, including both defect and non-defect cases: Alignment Defect, Barcode Defect, Good Sample, Expiration Date Defect, Item Code Defect, Item Name Defect, Receive Date Defect,

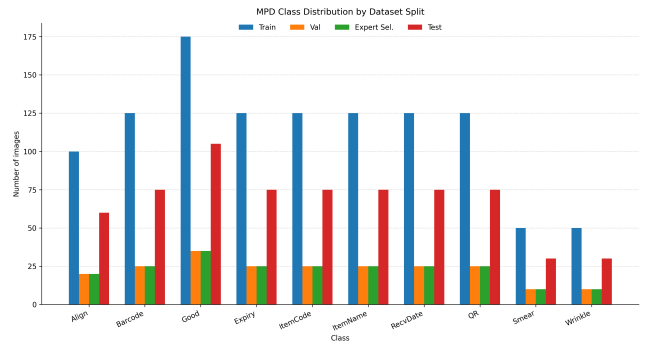


Fig. 5: Class-wise distribution of the Medical Packaging Dataset (MPD).

QR Defect, Smear Defect, and Wrinkle Defect. The class distribution reflects a typical industrial imbalance, where defect samples occur less frequently than normal samples. Specifically, the dataset contains 350 images of good samples, while defect categories range from 100 to 250 images per class, with Smear Defect and Wrinkle Defect being the least represented classes.

To ensure label consistency across all models and all stages of the workflow, we used a fixed class ID order for all experiments. This is important because the proposed framework combines outputs from multiple models and requires identical label mapping for correct fusion and decision-making. Medical packaging label inspection also contains practical challenges such as defect-scale variation and class-wise variability. In particular, visually confusing defects may have weak boundaries and may resemble background printing textures.

All images were annotated using bounding boxes in YOLO format with the annotation tool and subsequently verified through a double-checking process to ensure annotation correct. The dataset was randomly partitioned into training, validation, expert selection, and testing subsets using a 50% / 10% / 10% / 30% split, corresponding to 1,125 training images, 225 validation images, 225 expert selection images, and 675 test images. Random sampling was applied to reduce selection bias and to ensure that each subset properly represents the overall data distribution. The validation set was used for hyperparameter tuning and early stopping, the expert selection set was used for per class expert-model selection, and the test set was held out for the final HiMoE-F evaluation.

4.2 Evaluation Parameters

In this study, we evaluate both detection accuracy and inference latency. Detection accuracy is measured using standard object-detection metrics based on matching

predicted bounding boxes to ground-truth boxes using an IoU threshold. Inference latency is evaluated using mean per-image inference latency because the proposed HiMoE-F framework uses conditional expert execution and must maintain stable real-time performance.

A predicted box is treated as a true positive (TP) if it matches a ground-truth box of the same class with sufficient overlap. The overlap is measured by intersection-over-union (IoU), defined as

$$\text{IoU}(A, B) = \frac{|A \cap B|}{|A \cup B|}. \quad (1)$$

Using TP, false positives (FP), and false negatives (FN), we compute Precision and Recall as

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}, \quad (2)$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}. \quad (3)$$

We also report mean average precision (mAP) under two settings. First, mAP@0.5 evaluates average precision (AP) at IoU = 0.5. Second, we use COCO-style mAP@0.5:0.95, which averages AP over IoU thresholds from 0.5 to 0.95 with a step of 0.05:

$$\text{mAP}_{@0.5} = \text{AP}_{0.5}, \quad (4)$$

$$\text{mAP}_{@0.5:0.95} = \frac{1}{10} \sum_{k=0}^9 \text{AP}_{0.5+0.05k}. \quad (5)$$

In addition to accuracy metrics, we report mean per-image inference latency is measured around the full HiMoE-F inference call, including base inference, any expert inference executed for the frame, fusion (class-wise NMS), and the final decision output. For each model, the mean inference latency is computed over all evaluated images and reported in milliseconds.

To identify the most suitable model for each class, model performance was evaluated using a class-specialized expert selection strategy that ranks candidate models for each class using a weighted sum model that combines recall and inference latency. The motivation for this formulation is that larger models may achieve higher recall, but often at the cost of substantially higher inference latency. Therefore, selecting the best model based solely on recall may not yield the most practical solution for deployment.

In this work, the comparison was performed class-wise across candidate models. Let $R_{m,c}$ denote the recall of model m for class c . Since recall values may vary in scale across different classes, class-wise normalization

was first applied to ensure fair comparison among models within the same class. The normalized recall is defined as

$$R_{m,c}^{\text{norm}} = \frac{R_{m,c} - R_c^{\min}}{R_c^{\max} - R_c^{\min}}, \quad (6)$$

where $R_c^{\max}$ and $R_c^{\min}$ denote the maximum and minimum recall values obtained by all models for class c , respectively. If all models yield identical recall for a given class, the normalized recall value is assigned as 1 for all models in that class.

To account for computational efficiency, the average inference latency of each model was also incorporated. Let T_m denote the average inference latency of model m . Since lower latency is preferred, inverse min-max normalization was applied as

$$T_m^{\text{norm}} = \frac{T_m^{\max} - T_m}{T_m^{\max} - T_m^{\min}}, \quad (7)$$

where $T_m^{\max}$ and $T_m^{\min}$ represent the maximum and minimum latency values among all evaluated models. Under this formulation, faster models obtain higher normalized latency scores.

After normalization, a class-wise performance score was computed by combining recall and latency as follows:

$$S_{m,c} = 0.8R_{m,c}^{\text{norm}} + 0.2T_m^{\text{norm}}, \quad (8)$$

where a weight of 0.8 was assigned to recall and 0.2 to latency. This weighting reflects the priority of minimizing missed detections while still considering inference efficiency for practical implementation. In medical packaging inspection, missing a true defect may lead to a more serious consequence than a moderate increase in inference time. Therefore, recall was given higher importance during expert selection, while latency was still included to preserve real-time feasibility.

The overall performance score of each model was then obtained by averaging the class-wise scores over all C classes:

$$S_m = \frac{1}{C} \sum_{c=1}^C S_{m,c}. \quad (9)$$

A higher value of S_m indicates a more favorable trade-off between detection performance and computational cost. Accordingly, the final model for the expert selection stage was chosen as the model with the highest overall performance score under the defined evaluation setting.

4.3 Implementation and training settings

All experiments were conducted in a Linux environment using a dedicated Python conda environment. The hardware platform consisted of an NVIDIA GeForce RTX 2080 Ti graphics card with 11 GB of VRAM. Model training, validation, prediction, and inference latency reporting were performed using the Ultralytics framework (YOLO and RT-DETR implementations) with a PyTorch backend. Automated hyperparameter tuning was carried out using Optuna [2]. OpenCV was used for image I/O and visualization, and all inference outputs were saved as annotated images together with structured logs for later analysis.

We evaluated a set of real-time models included yolov3u, yolov5mu, yolov8m, yolov9m, yolov10m, yolov11m, yolov12m, and rt-detr-l. Each candidate model was tuned and trained under the same dataset split and evaluation protocol to ensure comparability.

For each model architecture, we created a separate Optuna study and ran 30 trials. In each trial, Optuna suggested a training configuration and the model was trained using the Ultralytics training pipeline. The optimization objective was validation recall. Training was run for a maximum of 100 epochs with early stopping patience of 10 epochs. Since GPU memory can be a limiting factor when exploring batch size and image-resolution combinations, GPU memory was managed between trials by releasing unused CUDA memory to avoid out-of-memory failures. After each study was completed, we selected the best trial and saved its trained weights together with the corresponding hyperparameter configuration. For consistency, mean latency was reported in milliseconds per image. Table 2 reports the validation performance of each model using its Optuna-selected best configuration, together with the selected hyperparameters.

4.4 Single-model comparison experiments

In this subsection, we benchmark all tuned candidate models on the expert selection split dataset to understand their detection performance and runtime behavior before constructing the HiMoE-F pipeline. All models are evaluated under the same dataset split and class-ID mapping, and we report both the accuracy metrics defined in Section 4.2 and the inference latency measurements using the same protocol. The models were evaluated under stricter inference conditions using a confidence threshold of 0.7 and an IoU threshold of 0.7 to retain only highly confident and accurately localized detections. After obtaining the Optuna-selected best

configuration for each model, we compared the overall performance of all eight models across all classes.

After benchmarking the tuned models, we further analyzed per class performance to identify which defect types benefit from specialization. Overall, most classes achieve near 100% performance across several models such as Alignment Defect, Expiration Date Defect, Item Code Defect, Item Name Defect, Receive Date Defect, QR Defect, and Wrinkle Defect are comparatively easy to detect under consistent label layouts. For these classes, YOLOv8m consistently achieves very high recall while also providing the best inference latency among the high-performing candidates.

In contrast, one class shows stronger detection to model choice is Barcode Defect benefits from a dedicated specialist. Among the evaluated models, YOLOv10m provides the highest weighted performance score for Barcode Defect (0.8958). Since HiMoE-F selects experts using the weighted combination of normalized recall and normalized latency, this result supports choosing YOLOv10m as the barcode specialist.

Following the rank strategy described in Section 4.2, we report the selected best model for each class in Table 3. Therefore, the final expert mapping selects YOLOv8m as the base expert detector and YOLOv10m as the class-specialized expert detector for Barcode Defect.

Although the absolute recall of YOLOv8m for Smear Defect is only 0.6923, this is still the highest recall achieved among all evaluated models for that class on the expert selection split. After weighted sum model, the recall score therefore becomes 1.0. Because YOLOv8m also has the lowest latency among the candidate models, the resulting weighted performance score ranks first for Smear Defect as well. This score reflects relative superiority among the evaluated models rather than perfect absolute detection performance.

4.5 Deployment evaluation of HiMoE-F

After the expert selection stage, YOLOv8m was selected as the base expert detector and YOLOv10m as the specialist for *Barcode Defect*. Although the framework supports multiple specialists, the final locked configuration selected one specialist class under the current dataset and scoring scheme. The final HiMoE-F pipeline was then evaluated on the 30% test split using the complete inference chain, including preprocessing, base expert detection, conditional expert execution, class-wise fusion, and final post-processing.

Table 4 compares the proposed HiMoE-F framework with the evaluated single-model baselines. The proposed

Table 2: Validation results on the 10% validation split using Optuna-selected best hyperparameter settings for each model.

Model	Recall	Precision	mAP@0.5	mAP@0.5:0.95	Optimizer	Image Size	Batch	lr ₀	Weight Decay
YOLOv3u	0.969	0.990	0.982	0.864	AdamW	512	8	0.0000117	0.0000168
YOLOv5mu	0.984	0.983	0.987	0.847	Adam	512	4	0.000198	0.00000498
YOLOv8m	0.981	0.982	0.983	0.865	AdamW	416	2	0.000103	0.001329
YOLOv9m	0.979	0.971	0.987	0.839	SGD	512	8	0.004128	0.004824
YOLOv10m	0.957	0.946	0.980	0.860	Adam	512	8	0.000376	0.000108
YOLOv11m	0.982	0.980	0.987	0.874	AdamW	512	4	0.000191	0.0000039
YOLOv12m	0.948	0.987	0.979	0.824	AdamW	640	1	0.000036	0.000192
RT-DETR-L	0.927	0.954	0.982	0.846	AdamW	512	4	0.000215	0.000086

Table 3: Best model selected for each class based on the proposed weighted performance score.

Class ID	Class	Best Model	Recall	Latency (ms)	Performance Score
0	Alignment Defect	YOLOv8m	1.0000	9.17	1.0000
1	Barcode Defect	YOLOv10m	0.9583	19.56	0.8958
2	Good Sample	YOLOv8m	1.0000	9.17	1.0000
3	Expiration Date Defect	YOLOv8m	1.0000	9.17	1.0000
4	Item Code Defect	YOLOv8m	1.0000	9.17	1.0000
5	Item Name Defect	YOLOv8m	1.0000	9.17	1.0000
6	Receive Date Defect	YOLOv8m	1.0000	9.17	1.0000
7	QR Defect	YOLOv8m	1.0000	9.17	1.0000
8	Smear Defect	YOLOv8m	0.6923	9.17	1.0000
9	Wrinkle Defect	YOLOv8m	1.0000	9.17	1.0000

method achieved the highest recall of 0.973, with a precision of 0.985, mAP@0.5 of 0.962, and mAP@0.5:0.95 of 0.84. These results confirm that the hierarchical expert strategy improves defect-sensitive detection performance in a recall-oriented industrial inspection setting.

In terms of inference latency, HiMoE-F required 43.34 ms per image on average, including preprocessing, inference, and post-processing. Although this is slower than the single-model baselines because of the additional cost of conditional expert execution, the average processing time still satisfies the real-time inspection requirement of less than 100 ms per image. Compared with YOLOv8m, which achieved 0.956 recall with 8.707 ms latency, HiMoE-F improves recall by 1.7 percentage points while remaining well below the practical deployment threshold of 100 ms per image.

It is also important to note that HiMoE-F does not outperform the strongest single-model baseline on every metric. In Table 4, YOLOv8m achieves slightly higher mAP@0.5 and mAP@0.5:0.95 than HiMoE-F. Therefore, the main advantage of the proposed framework is not universal metric dominance, but improved recall under a deployment-oriented design objective.

The class-wise performance of the final expert pipeline is reported in Table 5. Most categories achieved perfect recall. Specifically, *Alignment Defect*, *Good Sample*, *Expiration Date Defect*, *Item Code Defect*, *Item Name Defect*, *Receive Date Defect*, *QR Defect*, *Barcode Defect*, and *Wrinkle Defect* all reached a recall of 1.00. The most difficult category remained *Smear Defect*, which obtained 0.75 recall, 0.845 precision, 0.707 mAP@0.5,

and 0.444 mAP@0.5:0.95. This result indicates that smear-related defects remain the most challenging because of their irregular appearance, weak boundaries, and the presence of multiple objects in a single image.

The normalized confusion matrix in Fig. 6 further supports this observation. Most classes are concentrated on the main diagonal, indicating highly reliable classification and localization across defect categories. The main residual error pattern is associated with *Smear Defect*, where 25% of true smear instances are assigned to the background class. In addition, the background-confusion distribution is dominated by smear predictions. This suggests that visually confusing smear patterns remain difficult to separate from background regions and may also trigger false positives.

Table 4: Comparison of HiMoE-F with single-model baselines on the test split.

Model	Recall	Precision	mAP@0.5	mAP@0.5:0.95	Latency per image (ms)
YOLOv3u	0.953	0.991	0.969	0.869	18.351
YOLOv5mu	0.907	0.992	0.948	0.836	9.924
YOLOv8m	0.956	0.989	0.969	0.869	8.707
YOLOv9m	0.647	0.886	0.765	0.676	11.888
YOLOv10m	0.883	0.988	0.936	0.837	8.780
YOLOv11m	0.878	0.988	0.931	0.840	10.499
YOLOv12m	0.785	0.972	0.871	0.751	14.676
RT-DETR-L	0.862	0.935	0.909	0.809	19.735
HiMoE-F (Ours)	0.973	0.985	0.962	0.840	43.34

Table 5: Class-wise performance of the final HiMoE-F expert pipeline on the test split.

Class type	Precision	Recall	mAP@0.5	mAP@0.5:0.95
Alignment Defect	1.00	1.00	0.99	0.990
Barcode Defect	1.00	1.00	0.99	0.872
Good Sample	1.00	1.00	0.99	0.963
Expiration Date Defect	1.00	1.00	0.99	0.857
Item Code Defect	1.00	1.00	0.99	0.719
Item Name Defect	1.00	1.00	0.99	0.885
Receive Date Defect	1.00	1.00	0.99	0.883
QR Defect	1.00	1.00	0.99	0.859
Smear Defect	0.845	0.75	0.707	0.444
Wrinkle Defect	1.00	1.00	0.99	0.933

At the decision level, HiMoE-F demonstrated highly reliable practical behavior across the 675 test samples.

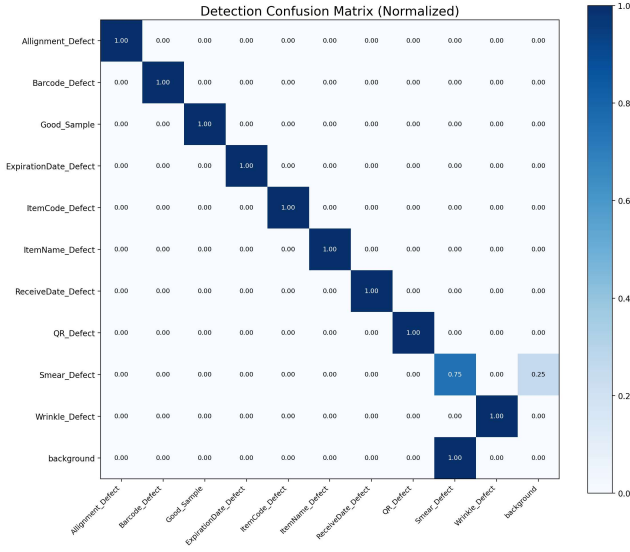


Fig. 6: Normalized confusion matrix of final HiMoE-F pipeline

The framework successfully identified all 118 ground-truth PASS samples as PASS, and correctly classified 554 out of 557 ground-truth REJECT samples as REJECT. Crucially, the model produced zero false acceptances; the remaining 3 REJECT samples were conservatively flagged as UNKNOWN. Fig. 7 shows representative detection results of the proposed HiMoE-F on the test set, demonstrating accurate localization and classification across a variety of defect types and visual conditions. These results confirm that the proposed framework provides a practical inspection strategy for automated quality control in medical packaging production, with improved defect sensitivity while preserving deployment-oriented efficiency.

5 Conclusion

Experimental results on the Medical Packaging Dataset (MPD) showed that HiMoE-F achieved the highest recall among all evaluated models, while maintaining average latency well below the industrial requirement of 100 ms per image. Compared with the strongest single-model baseline, the framework improved recall by 1.7 percentage points, although this gain came with higher latency and slightly lower mAP. These results indicate that the proposed approach provides a favorable trade-off for recall-oriented industrial inspection, where missed defects are typically more costly than moderate additional computation.

From an industrial perspective, the proposed framework offers a practical inspection strategy for automated

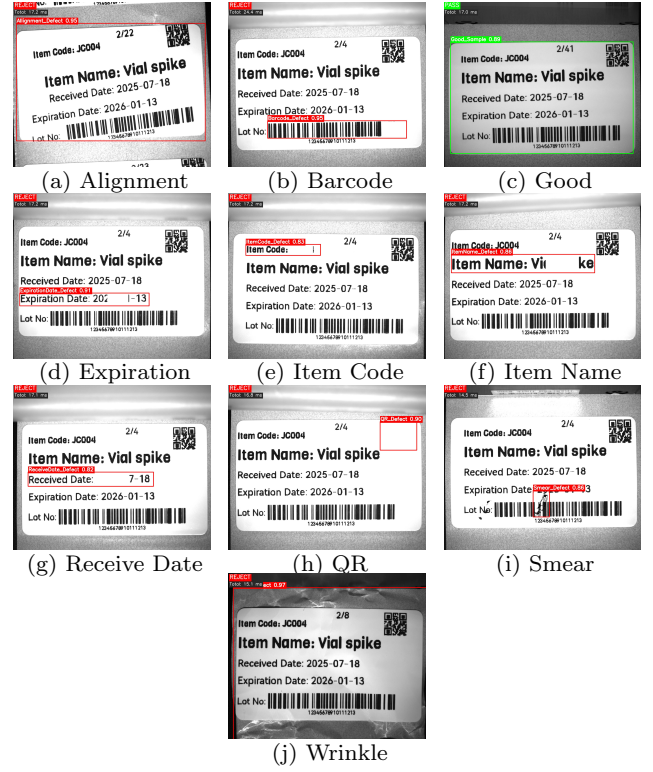


Fig. 7: Representative detection results of the proposed HiMoE-F on the test set.

quality control in medical packaging production. By combining class-specialized expert selection with trigger-based specialist execution, the system improves defect sensitivity while preserving deployment-oriented efficiency.

However, several limitations remain. The dataset was collected from a specific industrial environment, and some classes, especially Smear Defect, remain difficult to detect. The experiments were also conducted on a single hardware platform, which may limit direct generalization of the latency results. In addition, although the framework includes an UNKNOWN decision state for low-evidence deployment cases, true open-set performance was not evaluated because the current dataset does not contain a dedicated unknown class.

Future work will focus on improving performance for visually confusing defect classes, evaluating the framework on larger and more diverse industrial datasets, and testing the system in real production-line environments.

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Conflict of interest

The authors declare that they have no conflict of interest.

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